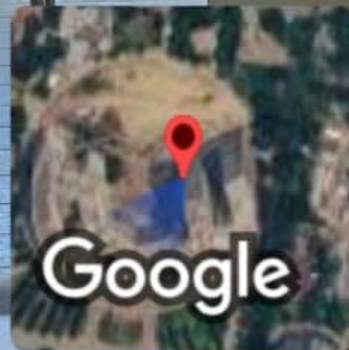




GPS Map Camera



Mevalurkuppam, Tamil Nadu, India



22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.026534° Long 80.016847°
Friday, 21/08/2026 09:40 AM GMT +05:30



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

AgriSphere AI – Intelligent Digital Agriculture Ecosystem

CSA0904-Programming in Java

Supervised By

Dr.R.Thalapathi Rajasekaran`

SIMATS Engineering

Presented By

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Introduction



- AgriSphere AI is an intelligent digital agriculture ecosystem that connects farmers, buyers, and service providers on a single platform.
- The platform enables farmers to create digital profiles and manage their agricultural products efficiently.
- Buyers can easily search, compare, and purchase agricultural products from trusted farmers.
- AI technology provides personalized recommendations based on user preferences and market trends.
- Smart notifications keep users informed about orders, payments, and important agricultural updates.
- Analytics dashboards help users monitor business performance and make informed decisions.
- The project promotes digital transformation by making agriculture smarter, faster, and more efficient.



PROBLEM STATEMENT



- Farmers often struggle to find genuine buyers and receive fair prices for their products.
- Buyers face difficulty in identifying trusted farmers and quality agricultural products.
- Agricultural services such as equipment rental and consultancy are not available on a single platform.
- Traditional farming and selling methods reduce efficiency and increase operational costs.
- Farmers have limited access to AI-based recommendations and real-time market insights.
- Existing agriculture applications do not provide complete digital solutions for all stakeholders.
- A centralized digital ecosystem is required to improve communication, sales, and agricultural productivity.



Project Scope & Objectives

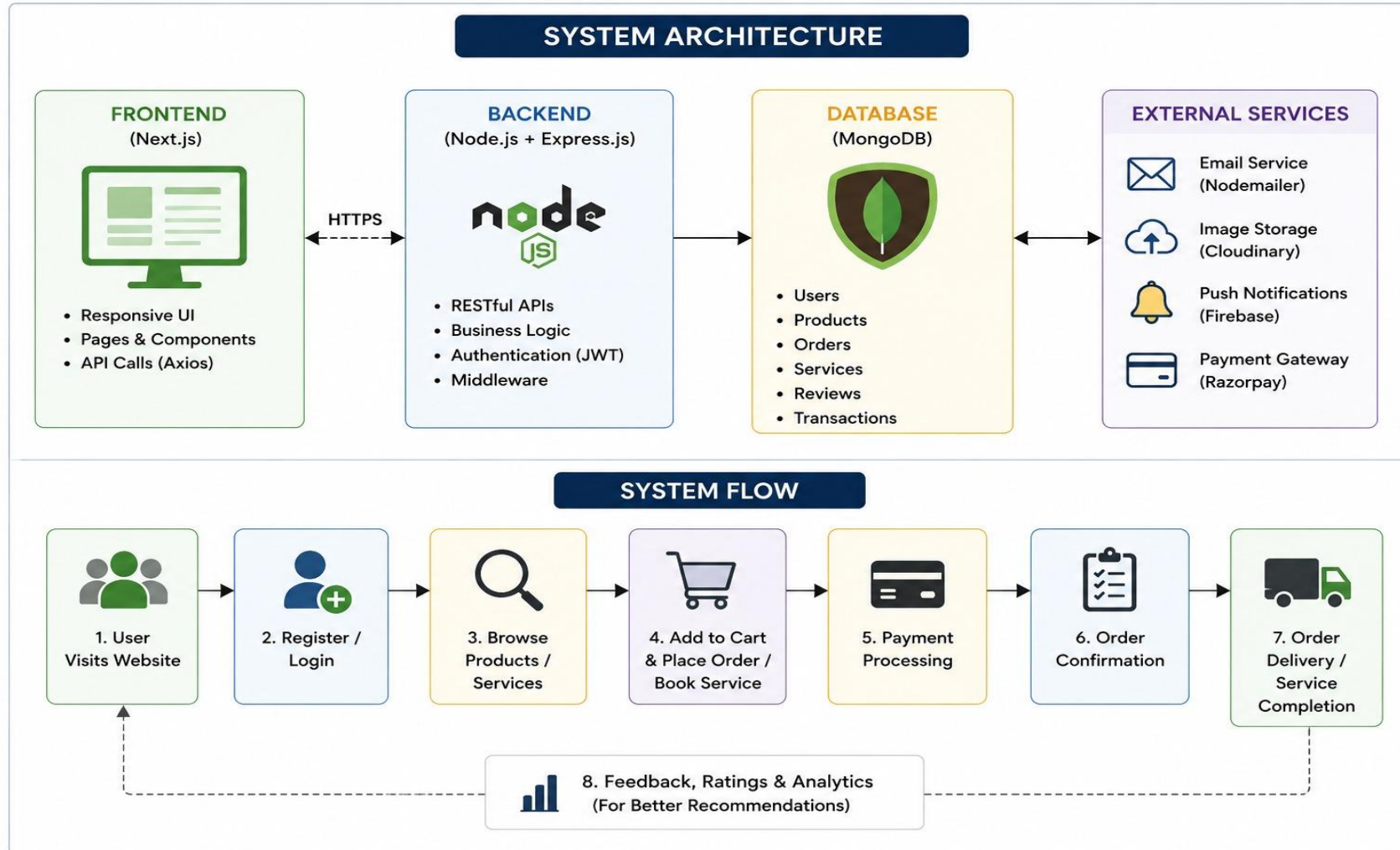


- Developed a centralized platform that connects farmers, buyers, service providers, and administrators.
- Allow farmers to manage products, receive orders, and track their sales digitally.
- Provide buyers with a secure marketplace to purchase agricultural products.
- Integrate AI to deliver personalized recommendations and market insights.
- Implemented smart notifications for orders, payments, and important updates.
- Generated analytics reports to help users monitor performance and business growth.
- Built a scalable system that can support future technologies like IoT and weather forecasting.

LITERATURE REVIEW

S. No	LITERATURE TITLE	AUTHOR AND YEAR	LIMITATIONS	HOW OUR PROJECT OVERCOMES IT
1	Smart Agriculture Monitoring Using IoT and Cloud Computing	Patel & Singh, 2022	Focuses only on farm monitoring and sensor data without providing an online marketplace.	Integrates smart farming with an AI-powered marketplace connecting farmers, buyers, and service providers.
2	AI-Based Crop Recommendation and Yield Prediction System	Kumar et al., 2023	Provides crop recommendations but does not support product selling or business management.	Combines AI recommendations with product management, buyer discovery, and business analytics.
3	Online Agricultural Marketplace for Farmers	Sharma & Verma, 2021	Supports online product selling but lacks AI recommendations and service provider integration.	Offers AI-based recommendations, service booking, smart notifications, and a complete digital agriculture ecosystem.
4	Digital Farming Platform with Precision Agriculture	Das & Roy, 2024	Concentrates on precision farming techniques without connecting buyers and agricultural services.	Connects farmers, buyers, and service providers while providing analytics, AI insights, and an integrated marketplace.

System Architecture





MODULES



Module 1 – User & Profile Management

Provides secure registration, login, and role-based access for farmers, buyers, service providers, and administrators. Users can create and manage their profiles within the platform.

Module 2 – Smart Digital Marketplace

Enables farmers to list agricultural products while buyers can search, compare, and purchase them through an easy-to-use online marketplace.

Module 3 – AI Recommendation Engine

Uses user preferences and browsing history to provide personalized product recommendations, improving product discovery and customer experience.

Module 4 – Order & Service Management

Handles shopping cart operations, order placement, payment processing, service bookings, and real-time order tracking for users.

Module 5 – Analytics & Administration

Provides administrators with dashboards to monitor users, products, orders, and sales while generating reports and business insights for better decision-making.



Module 1 – User & Profile Management



- ▶ Provides secure registration and login functionality for farmers, buyers, service providers, and administrators, with different access permissions based on the user's role.
- ▶ Allows farmers and service providers to create and maintain digital profiles containing their relevant details, enabling buyers to identify and connect with trusted users.
- ▶ Implements role-based access control so that each type of user can access only the features and operations relevant to their responsibilities.
- ▶ Enables users to manage their account information while maintaining secure authentication and controlled access to marketplace, services, orders, and administrative features.
- ▶ Creates the basic user-management layer that connects all stakeholders within the centralized AgriSphere AI digital agriculture ecosystem.



Engineer to Excel

SAMPLE OUTPUT



 Marketplace

 Agri Services

 AI Advisory 



Sign In

Register

I WANT TO JOIN AS



Farmer

Sell crops & book services



Buyer

Buy fresh farm produce



Ag-Seller

Sell inputs & equipment



Services

Offer farm services

First Name *

 e.g. Ramesh

Last Name *

 e.g. Patel

Email Address *

 name@domain.com

Phone Number

 +91 98765 43210

Location / District *

 e.g. Nashik, Maharashtra

Password *

 Min. 6 characters



Confirm Password *

 Re-enter password



 FARM DETAILS

localhost:3000/products



Module 2 – Smart Digital Marketplace



- ▶ Provides farmers with a digital marketplace where they can list, manage, and sell agricultural products directly to potential buyers without depending entirely on traditional selling methods.
- ▶ Allows buyers to browse, search, and compare agricultural products based on the available product information before making purchasing decisions.
- ▶ Creates a direct connection between farmers and genuine buyers, helping farmers improve product reach while allowing buyers to access products from trusted agricultural sources.
- ▶ Integrates product discovery and purchasing into a single online platform, reducing the difficulty of finding agricultural products and improving the overall buying and selling process.
- ▶ Supports the project's objective of creating a centralized marketplace that combines agricultural product management, buyer discovery, and digital sales within the same ecosystem.

SAMPLE OUTPUT



**AgriSphere**
FARM TO FORK & ADVISORY

Marketplace

Agri Services

AI Advisory 

 Sign In [Register](#)

 Filters

Reset

Organic Certified Only ☐

CATEGORIES

All Categories

Fresh Vegetables

Fresh Fruits

Grains, Wheat & Rice

Bio & NPK Fertilizers

Certified Seeds

Max Price

₹50

₹20,000

₹20,000+

Showing 20 items

Sort: **Featured / Best Match**

Organic

PRODUCE



Fresh Vegetables

AgriSphere Marketplace

650 kg left

4.9 (42)

Nashik, Maharashtra

Fresh Farm Organic Vine Tomatoes

By Ramesh Patel

Price

₹38 / kg

Add

Organic

PRODUCE



Fresh Vegetables

AgriSphere Marketplace

2500 kg left

4.7 (22)

Nashik, Maharashtra

Crisp Organic Nashik Red Onions

By Ramesh Patel

Price

₹26 / kg

Add

Organic

PRODUCE



Fresh Vegetables

AgriSphere Marketplace

300 kg left

4.8 (18)

Nashik, Maharashtra

Premium Farm Fresh Spinach (Palak) Bunch

By Ramesh Patel

Price

₹22 / kg

Add

PRODUCE



Organic

PRODUCE



Organic

PRODUCE





Module 3 – AI Recommendation Engine



- ▶ Uses user preferences and browsing behaviour to understand the interests of buyers and generate personalized agricultural product recommendations.
- ▶ Analyzes user interaction with available products to improve product discovery, helping users find relevant products instead of manually searching through the entire marketplace.
- ▶ Provides AI-based recommendations that can support users in making better farming and business decisions based on their requirements and available marketplace information.
- ▶ Integrates AI capabilities with the marketplace so that recommendations are not isolated features but directly contribute to product discovery and customer experience.
- ▶ Extends the platform beyond a traditional agricultural marketplace by combining AI recommendations, marketplace data, and business insights in one digital ecosystem.



Engineer to Excel

Sample Output



AgriSphere
FARM TO FORK & ADVISORY

Marketplace

Agri Services

AI Advisory AI



Sign In

Register

Scientific Agronomy & Soil Intelligence

AI Crop & Soil Health Advisory Hub

Tailored agronomic recommendations based on your soil type, measured pH level, and crop cycle. Maximize harvest yield while cutting chemical fertilizer wastage by up to 35%.

Farm Parameters Profile Default

Soil Classification

Black Cotton Soil (Heavy Clay)

Soil pH Level 6.8

Optimal Neutral (Ideal for most crops)

Active / Planned Crop

Tomato (Hybrid Vine)

Current Growth Stage

Flowering & Fruit Setting

Season

Kharif (Monsoon)

Acreage

5

Re-Calculate Advisory

LIVE DIAGNOSTICS

Tomato on Black Cotton Soil (5 Acres)

Yield Potential: 94%

Based on your soil pH of 6.8 during the **Flowering & Fruit Set** stage, phosphorus uptake is optimal, while foliar zinc and potassium supplementation will prevent flower drop and enhance fruit weight by an estimated 18-24%.



CROP ADVISORY • SOIL NITROGEN REPLENISHMENT BASED ON YOUR BLACK COTTON SOIL PROFILE & PH

6.8

Optimal Kharif Season Crop Rotation: Soybean to Gram

96% Match

Following your harvest of Soybean, cultivating Gram (Chickpea) will naturally fix 35-40 kg of atmospheric nitrogen per hectare into your soil, lowering future fertilizer costs by 22%.

ACTION STEPS:

- Optimal sowing window: October 15 – November 10
- Seed rate: 65-75 kg/ha with Rhizobium bio-inoculant treatment
- Yield potential: 18-22 quintals/acre under drip irrigation

Directly Recommended Inputs:



Module 4 – Order & Service Management



- ▶ Manages the complete transaction flow from adding products to the shopping cart and placing an order to payment processing and order confirmation.
- ▶ Provides support for agricultural service bookings, allowing users to access services offered by service providers through the same centralized platform.
- ▶ Maintains order status and provides real-time tracking, helping users monitor the progress of their purchases and reducing uncertainty during delivery.
- ▶ Integrates payment processing with order management so that transactions, confirmations, and related notifications can be handled as part of the same workflow.
- ▶ Supports smart notifications for orders, payments, and important transaction updates, improving communication between users and reducing delays in the overall process.

Shopping Cart (3 items)

Review and adjust items before proceeding to secure checkout

[Clear Cart](#)



Fresh Farm Organic Vine Tomatoes
₹38 / kg

— 1 +

₹38



Crisp Organic Nashik Red Onions
₹26 / kg

— 1 +

₹26



Alphonso Mangoes — Devgad GI Tag (Box of 12)
₹850 / box (12 pcs)

— 1 +

₹850



Order Summary

 COUPON CODE (E.G. KISAN10)

Apply

Subtotal

₹914

Agri Shipping

₹120

Total Amount

₹1,034

Proceed to Checkout →



Module 5 – Analytics & Administration



- ▶ Provides administrators with centralized dashboards to monitor registered users, agricultural products, orders, transactions, and overall sales activity across the platform.
- ▶ Generates analytical reports that help administrators understand business performance, sales trends, user activity, and platform operations for better decision-making.
- ▶ Enables monitoring of the marketplace and its different stakeholders, helping administrators maintain the reliability, security, and efficient operation of the digital ecosystem.
- ▶ Uses collected platform data to provide business insights and performance analysis, supporting farmers and administrators in understanding growth and transaction patterns.
- ▶ Provides the management and analytics layer required to make AgriSphere AI a scalable digital agriculture platform rather than only an online product-selling application.



Engineer to Excel

SAMPLE OUTPUT



**AgriSphere**
FARM TO FORK & ADVISORY

[Marketplace](#)

[Agri Services](#)

[AI Advisory](#) 

 3

[Dashboard](#)

 M 

 **P**

Verified Produce Buyer

Priya Sharma

 Pune, Maharashtra • Restaurant & Wholesale Sourcing

[Browse Fresh Farm Produce](#)

Total Sourced Value



₹38,400

Direct Farm Purchases

Active Shipments



1 Order

In Transit to Baner Road

Delivered Orders



4 Orders

100% Quality Fulfilled

Favorite Farms



3 Farms

Nashik Organic Partners

Recent Orders & Deliveries

[View All Orders](#)

AGR-2024-8841

150x Fresh Farm Organic Vine Tomatoes, 50x Premium Aged Traditional Basmati Rice

₹11,400

Processing

[Live Tracking →](#)

AGR-2024-7729

200x Crisp Organic Nashik Red Onions

₹5,200

Completed

[Live Tracking →](#)

Saved Direct Farm Partners

Green Valley Agro Farm

Nashik, MH • Vine Tomatoes & Onions

Organic

Sahyadri Pomegranate Orchards

Sangamner, MH • Bhagwa Variety

Export Grade

[Find More Verified Farms](#)

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Result and Outcome



- Farmers can directly connect with buyers through the digital marketplace, providing better access to customers and supporting increased income through digital product sales.
- Buyers can search, compare, and purchase agricultural products from trusted farmers through a centralized platform, making the overall buying process easier and more efficient.
- Service providers can offer services through the same platform, creating a common digital ecosystem where users can access agricultural products as well as related services.
- The AI recommendation engine provides personalized product recommendations based on user preferences and interactions, helping users discover relevant products and make better farming and business decisions.
- The project brings farmers, buyers, service providers, and administrators together on a single platform, promoting a smarter, more efficient, and technology-driven agricultural ecosystem.



Future Scope and recommendations

- **IoT-Based Smart Farming:** Integrate IoT sensors to collect real-time data such as soil moisture, temperature, humidity, and crop conditions, enabling smarter farm monitoring.
- **Advanced AI Recommendations:** Enhance the AI system to provide more advanced recommendations for crops, products, farming activities, and market decisions using larger amounts of user and agricultural data.
- **Real-Time Market Insights:** Integrate live agricultural market data to provide farmers with current price trends and demand information, helping them decide when and where to sell their products.
- **Advanced Predictive Analytics:** Extend the analytics module with predictive models for sales, demand, product trends, and business growth to support proactive decision-making.
- **Mobile Application Development:** Extend the platform into dedicated mobile applications so farmers, buyers, and service providers can access marketplace, orders, services, and notifications conveniently from mobile devices.



Conclusion



- AgriSphere AI is a complete digital agriculture ecosystem.
- It connects farmers, buyers, and service providers in one platform.
- AI improves efficiency through smart recommendations and analytics.
- The marketplace simplifies agricultural buying and selling.
- Multiple user roles make the system flexible and scalable.
- The project supports future expansion with advanced AI features.
- AgriSphere AI contributes to modern, smart, and sustainable agriculture.